

Question

There is currently an alloy primarily composed of Pb and Zn, which may contain other metals. To determine the content of each component, an experimenter designed the following experimental procedure:

(1) Dissolve the 0.7800 g sample in an excess mixed acid, dilute to 250.0 mL, and transfer 25.00 mL of the solution to a conical flask. Add 0.1 g NaOH and 10 mL of 1 : 1 ammonia buffer solution, followed by a small amount of tartaric acid and 0.2 g KCN. Using Eriochrome Black T as the indicator, titrate with a standard EDTA solution of concentration c_1 ($\text{mol} \cdot \text{L}^{-1}$). When the solution changes from purple-red to blue, record the reading V_1 (mL). Then, take another 25.00 mL of the solution, add NaOH and ammonia buffer solution, and titrate with the EDTA solution, recording the reading V_2 (mL).

In the first experiment, the experimenter prepared a standard EDTA solution with $c_1 = 0.02016$ ($\text{mol} \cdot \text{L}^{-1}$). When the titration volume reached 17.52 mL, he realized there was an issue with the experiment and abandoned it.

In light of this, he designed a second experimental procedure:

(2) Subsequently, the experimenter revised the procedure as follows: Dissolve the 0.7800 g sample in an excess mixed acid, dilute to 250.0 mL, and transfer 25.00 mL of the solution to a conical flask. Add 0.1 g NaOH and 10 mL of 1 : 1 ammonia buffer solution, followed by a small amount of tartaric acid and 0.2 g KCN. Then, introduce 0.2 g of triethanolamine, and using Eriochrome Black T as the indicator, titrate with an EDTA solution of concentration c_1 ($\text{mol} \cdot \text{L}^{-1}$) until the endpoint is reached, recording the reading V_3 (mL). Next, add a slight excess of 1 : 1 formaldehyde solution and continue titrating until the solution just turns blue, recording the reading V_4 (L).

Regarding the above analysis process, which of the following statements is correct? (For options involving calculations, an error within 1% is considered acceptable.)

A. In Scheme 1, the mass fraction of Pb can be expressed as $2.478c_1V_1$

B. In Scheme 1, the mass fraction of Zn can be expressed as $-0.8583c_1(V_1 - V_2)$.

- C.** The experimenter believes that the reason for the failure of Scheme 1 is that the solution turned blue prematurely when titrated to 17.52 mL.
- D.** In Scheme 2, when formaldehyde is added, the sum of the coefficients on the product side of the ionic equation for the reaction of formaldehyde (with the coefficients in the simplest whole-number ratio) is 8.
- E.** In Scheme 2, replacing triethanolamine with ascorbic acid + potassium cyanide will not affect the titration process.
- F.** If the sample contains the metal cadmium, according to Method 2, cadmium will block the indicator and affect the measurement results.
- G.** In Scheme 2, the mass fraction of Zn can be expressed as $0.8983c_1(V_4 - V_3)$.
- H.** None of the above statements are correct

Answer

Correct Answer: **H**

Detailed Explanation

Titration process of Method 1:

After adding potassium cyanide, Zn^{2+} forms $\text{Zn}(\text{CN})_4^{2-}$ and is masked. EDTA only titrates Pb^{2+} , with Eriochrome Black T as the indicator. The solution changes from purple-red to blue. Under an ammonia buffer solution, both Zn^{2+} and Pb^{2+} are titrated by EDTA.

CHECKPOINT

1 PTS

After adding potassium cyanide, Zn^{2+} forms $\text{Zn}(\text{CN})_4^{2-}$ and is masked

Thus, V_1 represents the amount of Pb^{2+} , and V_2 represents the total amount of Zn^{2+} and Pb^{2+} .

Therefore, the amount of Zn^{2+} is $V_2 - V_1$.

CHECKPOINT

1 PTS

V_1 represents the amount of Pb^{2+} , and the amount of Zn^{2+} is $V_2 - V_1$

The mass fraction expression for Pb should be:

$$\frac{c_1 V_1 \times 207.2 \times 10}{0.7800 \times 1000} = 2.6564 c_1 V_1.$$

Option A's expression is incorrect.

CHECKPOINT

1 PTS

The mass fraction of Pb is $2.6564c_1V_1$, and Option A is incorrect

The mass fraction of Zn is:

$$\frac{c_1(V_2 - V_1) \times 65.38 \times 10}{0.7800 \times 1000} = 0.8383c_1(V_2 - V_1).$$

Option B's expression is incorrect.

CHECKPOINT

1 PTS

The mass fraction of Zn is $0.8383c_1(V_2 - V_1)$, and Option B is incorrect

Analysis of the first failed experiment: For the first titration of lead content, assuming the sample consists entirely of lead, the theoretical endpoint should be $0.7800/207.2/0.02016 \times 1000 = 18.67 \text{ mL}$. The experimenter titrated up to 17.52 mL , only about 1 mL short of the endpoint. At this point, each drop of titrant added should cause localized color change to the endpoint color due to excessive local concentration, reverting to the pre-endpoint color upon mixing. The experimenter likely observed no color change at all and concluded the experiment failed, so Option C is incorrect. In this case, the failure was caused by indicator僵化.

CHECKPOINT

0.5 PTS

Excessive local concentration causes localized color change to the endpoint color, reverting to the pre-endpoint color upon mixing

CHECKPOINT

1 PTS

No color change was observed, leading to the conclusion that the experiment failed, so Option C is incorrect.

CHECKPOINT

0.5 PTS

The failure was caused by indicator僵化

Method 2 differs from Method 1 by adding triethanolamine, which is commonly used to mask Fe^{3+} , Al^{3+} , thereby preventing 僵化 of Eriochrome Black T. Since the failed experiment cannot determine which ion caused the indicator僵化, the possibility of both Fe^{3+} , Al^{3+} being present must be considered. Therefore, triethanolamine is added instead of the ascorbic acid + potassium cyanide combination, which only masks Fe^{3+} , making Option E incorrect.

CHECKPOINT

0.5 PTS

Triethanolamine is used to mask Fe^{3+} , Al^{3+}

CHECKPOINT

0.5 PTS

The possibility of both Fe^{3+} , Al^{3+} being present must be considered

CHECKPOINT

1 PTS

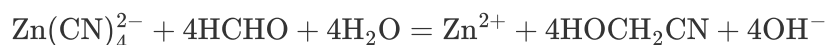
Triethanolamine is added instead of the ascorbic acid + potassium cyanide combination, which only masks Fe^{3+} , making Option E incorrect

The purpose of adding formaldehyde is to demask the masked $\text{Zn}(\text{CN})_4^{2-}$ for continuous titration. The ionic equation is:

CHECKPOINT

1 PTS

Formaldehyde demasks $\text{Zn}(\text{CN})_4^{2-}$



The sum of the product coefficients is 9, making Option D incorrect.

CHECKPOINT

1 PTS

$\text{Zn}(\text{CN})_4^{2-} + 4\text{HCHO} + 4\text{H}_2\text{O} = \text{Zn}^{2+} + 4\text{HOCH}_2\text{CN} + 4\text{OH}^-$ The sum of the product coefficients is 9, making Option D incorrect.

Cd^{2+} formed from metallic cadmium is masked by the added potassium cyanide and does not block the indicator, making Option F incorrect.

CHECKPOINT

1 PTS

Cd^{2+} is masked by the added potassium cyanide and does not block the indicator, making Option F incorrect.

Method 2 follows the same principle as Method 1, involving masking followed by back-titration. Thus, the volume of titrant corresponding to Zn^{2+} is $V_4 - V_3$, and the mass fraction is $0.8383c_1(V_4 - V_3)$, making Option G incorrect.

CHECKPOINT

1 PTS

The mass fraction of Zn is $0.8383c_1(V_4 - V_3)$, making Option G incorrect.

In conclusion, Options A-G are all incorrect, and Option H is correct.

CHECKPOINT

1 PTS

Options A-G are all incorrect, and Option H is correct.